

The Gaussian Moments Conjecture: Dimension Two and Sparse Minimality in Dimension Three

Roy van Rijn
Independent researcher

24 July 2026

Abstract

The Gaussian Moments Conjecture asks whether a polynomial whose positive powers all have Gaussian integral zero must eventually remain zero after multiplication by any fixed polynomial. We prove the conjecture for two real Gaussian variables. The proof uses circular coordinates to separate rotation from radius. The angular part becomes a constant-term problem, while the radial part becomes the simple functional $U^j \mapsto j!$. An exposed lower face of the resulting Newton diagram, the constant-term theorem of Duistermaat and van der Kallen, and reduction modulo a prime then isolate a nonzero lowest term. This forces every polynomial with vanishing pure moments to have rotational weights of one strict sign, from which the mixed-moment conclusion is immediate. Together with the recent counterexamples of Long in every dimension at least three, this determines the exact dimensional range of the conjecture. We then study minimal failure in the first failing dimension. An exhaustive exact search over all 59,535 supports having degree at most four and at most four monomials shows that every pure-moment-null polynomial in this range still satisfies the conjectural mixed-moment conclusion. Thus Long's five-term quartic is Pareto-minimal with respect to degree and circular-coordinate support size. The sparse theorem is computer-assisted: saturated moment ideals through order twenty-four are computed over $\mathbb{Q}$, and the three genuine surviving families are proved harmless at all orders.

1 Introduction

Let $X_1, \dots, X_d$ be independent standard real Gaussian random variables. The Gaussian Moments Conjecture, introduced by Derksen, van den Essen, and Zhao [1], is the following statement.

Gaussian Moments Conjecture GMC(d). If $P \in \mathbb{C}[X_1, \dots, X_d]$ satisfies

$$\mathbb{E}(P^m) = 0 \quad \text{for every } m \geq 1,$$

then, for every $Q \in \mathbb{C}[X_1, \dots, X_d]$,

$$\mathbb{E}(QP^m) = 0 \quad \text{for all sufficiently large } m.$$

The conjecture was motivated in part by its connection with the Jacobian Conjecture [1]. Long recently found explicit counterexamples to GMC(d) for every $d \geq 3$ [3]. The two-variable case was therefore the last undecided dimension.

Theorem 1.1. *The Gaussian Moments Conjecture holds in two real variables.*

Long's three-variable witness, in circular coordinates W, Z and an independent real Gaussian T , is

$$P_L = W + WZ - T^2 - \frac{3}{2}ZT^2 - \frac{1}{2}Z^2T^2. \quad (1)$$

It has degree four and five monomials, all its pure moments vanish, and $\mathbb{E}(ZP_L^m) = m!$ for every $m \geq 1$ [3]. Our second main result places this witness at the first corner of the sparse degree–support plane.

Theorem 1.2 (Sparse Pareto minimality). *Let W, Z, T be Gaussian Witt coordinates, so that*

$$\mathbb{E}(W^i Z^j T^k) = \begin{cases} i!(k-1)!!, & i = j \text{ and } k \text{ is even,} \\ 0, & \text{otherwise.} \end{cases}$$

Suppose that $P \in \mathbb{C}[W, Z, T]$ has total degree at most four and at most four nonzero monomials. If $\mathbb{E}(P^m) = 0$ for every $m \geq 1$, then, for every $Q \in \mathbb{C}[W, Z, T]$,

$$\mathbb{E}(QP^m) = 0 \quad \text{for all sufficiently large } m.$$

Consequently no GMC counterexample has both degree at most four and support size at most four in a Gaussian Witt frame.

In particular, (1) is minimal for the product partial order on $(\deg P, \# \text{Supp}(P))$. The theorem does not separately prove that degree four is globally minimal: a denser cubic is not excluded. Nor does it exclude a four-term counterexample of degree greater than four.

The proof is short once the right coordinates are chosen. In two dimensions, rotation has a single integer weight and there is a single radial variable. Thus every polynomial can be displayed schematically as

$$P = \sum_k \underbrace{T^k}_{\text{rotation}} \underbrace{B_k(U)}_{\text{radius}}.$$

Gaussian integration first takes the constant term in T , then replaces U^j by $j!$. If both positive and negative rotational weights occur, the lowest radial orders have a supporting line above weight zero. The polynomial carried by that lower face has a nonzero constant term in some power, by a theorem of Duistermaat and van der Kallen [2]. Raising that power to a suitable prime and reducing modulo the prime makes the lowest term impossible to cancel after the factorial substitution. This contradicts the assumed moment vanishing.

The argument is conceptual rather than computational. In particular, it contains the previously studied three-level and star-shaped supports without needing separate case analyses.

2 Circular coordinates

Let X, Y be independent standard real Gaussian variables. Set

$$Z = \frac{X + iY}{\sqrt{2}}, \quad W = \frac{X - iY}{\sqrt{2}}.$$

Wick's formula gives

$$\mathbb{E}(Z^a W^b) = \begin{cases} a!, & a = b, \\ 0, & a \neq b. \end{cases} \quad (2)$$

Introduce a radial variable U and a Laurent variable T through the injective substitution

$$Z \mapsto T, \quad W \mapsto UT^{-1}.$$

Thus $Z^a W^b$ becomes $T^{a-b} U^b$. Every polynomial $P(Z, W)$ therefore has a unique expression

$$P = \sum_{k \in S} T^k B_k(U), \quad (3)$$

where $S \subset \mathbb{Z}$ is finite and every $B_k \in \mathbb{C}[U]$ is nonzero. The integer k is the rotational weight.

Define the factorial functional

$$\mathcal{L} : \mathbb{C}[U] \longrightarrow \mathbb{C}, \quad \mathcal{L}(U^j) = j!.$$

Equation (2) is exactly the identity

$$\mathbb{E}(F) = \mathcal{L}(\text{CT}_T F) \quad (F \in \mathbb{C}[Z, W]). \quad (4)$$

This is the feature special to two variables: there is one angular constant-term operation and one radial factorial.

3 The exposed lower face

For each $k \in S$, write

$$\nu_k = \text{ord}_U B_k, \quad b_k = [U^{\nu_k}] B_k \neq 0.$$

The points (k, ν_k) record the smallest radial degree at each rotational weight. We now extract the lower face visible from weight zero.

Lemma 3.1 (Lower face). *Suppose $0 \in \text{conv}(S)$. There are rational numbers ρ, θ , a subset*

$$S_0 = \{k \in S : \nu_k = \rho + \theta k\},$$

and an integer $r \geq 1$ with the following properties:

$$\nu_k \geq \rho + \theta k \quad (k \in S), \quad 0 \in \text{conv}(S_0),$$

and, for

$$P_0(T) = \sum_{k \in S_0} b_k T^k, \quad F_m(U) = \text{CT}_T(P^m),$$

there are $n \in \mathbb{Z}_{\geq 0}$ and $c \in \mathbb{C}^\times$ such that

$$\text{every term of } F_m \text{ has degree at least } \rho m, \quad (5)$$

$$n = \rho r, \quad F_r(U) = cU^n + \text{terms of degree greater than } n. \quad (6)$$

Proof. Minimize

$$\sum_{k \in S} \lambda_k \nu_k$$

subject to

$$\lambda_k \geq 0, \quad \sum_k \lambda_k = 1, \quad \sum_k k \lambda_k = 0.$$

The feasible set is nonempty because $0 \in \text{conv}(S)$. Let the minimum be ρ . Elementary linear-programming duality supplies $\theta \in \mathbb{Q}$ such that

$$\nu_k \geq \rho + \theta k.$$

An optimal solution is supported on the equality set S_0 , so $0 \in \text{conv}(S_0)$.

We use the constant-term theorem of Duistermaat and van der Kallen [2]: if every positive power of a complex Laurent polynomial has zero constant term, then zero does not lie in the convex hull of its support. Applied contrapositively to P_0 , it gives an $r \geq 1$ for which

$$c := \text{CT}_T(P_0^r) \neq 0.$$

Consider a weight-zero product of m terms selected from (3). If the selected weights are $k_1, \dots, k_m$, then its radial degree is at least

$$\sum_{a=1}^m \nu_{k_a} \geq \rho m + \theta \sum_{a=1}^m k_a = \rho m.$$

This proves (5). When $m = r$, equality can occur only by taking the lowest coefficient from weights on S_0 , and the sum of all equality terms is c . Hence the lowest term of F_r is $cU^{\rho r}$. Its exponent is necessarily a nonnegative integer; call it n . $\square$

4 Prime isolation

The next lemma is the arithmetic step. Its role is simple: after a prime dilation, Frobenius preserves only p -th powers, while the factorial functional kills every radial level above the lowest one.

Lemma 4.1 (Prime-factorial isolation). *Let P and F_m be as in Lemma 3.1. If*

$$\mathcal{L}(F_m) = 0 \quad (m \geq 1),$$

then the conclusion (6) is impossible.

Proof. Let $R \subset \mathbb{C}$ be the $\mathbb{Z}$ -algebra generated by the coefficients of P and by c^{-1} . It is a finitely generated domain of characteristic zero. For all but finitely many rational primes p , there is a homomorphism from R to a field of characteristic p . Indeed, the finite-type morphism $\text{Spec } R \rightarrow \text{Spec } \mathbb{Z}$ is dominant, and its image is constructible by Chevalley's theorem [4]. Since it contains the generic point, it contains a nonempty open subset [5]. Because c is invertible in R , its image remains nonzero under every such homomorphism.

Fix one of these primes. Before reducing modulo p , work in the characteristic-zero domain R . Lemma 3.1 says that every term of F_{rp} has degree at least $\rho rp = np$. Hence $\mathcal{L}(F_{rp}) = 0$ can be written in R as

$$0 = (np)! \sum_{d \geq np} [U^d] F_{rp} \frac{d!}{(np)!}.$$

Since R is a domain of characteristic zero, $(np)! \neq 0$, so it may be cancelled to give

$$\sum_{d \geq np} [U^d] F_{rp} \frac{d!}{(np)!} = 0. \tag{7}$$

We now apply a homomorphism $R \rightarrow \kappa$, where κ is a field of characteristic p , and continue to use the same notation for the reduced polynomials.

In characteristic p , Frobenius and constant-term extraction give

$$F_{rp} = \text{CT}_T((P^r)^p) = (\text{CT}_T(P^r))^p = F_r^p. \quad (8)$$

Indeed, if $H = \sum_i h_i(U)T^i$, then $H^p = \sum_i h_i(U)^p T^{pi}$ in characteristic p ; because $p > 0$, the equality $pi = 0$ holds exactly when $i = 0$. Thus $\text{CT}_T(H^p) = \text{CT}_T(H)^p$. By (6), the coefficient of U^{np} in (8) is c^p . A term whose degree is not divisible by p has zero coefficient modulo p . A remaining term has degree jp for some $j > n$, and its multiplier in (7) is

$$\frac{(jp)!}{(np)!}.$$

This integer is divisible by p , since its product contains $(n+1)p$. Consequently, reducing (7) modulo p leaves only

$$c^p = 0,$$

contrary to the choice of the reduction. □

5 Proof of the theorem

Proof of Theorem 1.1. Write P as in (3) and suppose that $\mathbb{E}(P^m) = 0$ for every $m \geq 1$. By (4), this says

$$\mathcal{L}(F_m) = 0 \quad (m \geq 1).$$

If $0 \in \text{conv}(S)$, Lemma 3.1 produces a nonzero lowest coefficient and Lemma 4.1 gives a contradiction. Therefore

$$0 \notin \text{conv}(S).$$

Since $S \subset \mathbb{Z}$, either every weight of P is strictly positive or every weight is strictly negative.

Now fix a polynomial Q . Its rotational weights form a bounded finite set. Every weight of P^m , on the other hand, tends uniformly toward the same strict side as $m \rightarrow \infty$. Hence QP^m has no weight-zero term once m is sufficiently large:

$$\text{CT}_T(QP^m) = 0.$$

Equation (4) then gives $\mathbb{E}(QP^m) = 0$, as required. □

Corollary 5.1. *The Gaussian Moments Conjecture holds in dimension d exactly when $d \leq 2$.*

Proof. The case $d = 1$ is known [1, Proposition 4.2]. Theorem 1.1 proves the case $d = 2$. Long's examples disprove the conjecture for every $d \geq 3$ [3]. □

6 Sparse minimality in the first failing dimension

We prove Theorem 1.2. The finite part is an exhaustive exact coefficient-scheme calculation; the all-order interpretation of its survivors is conceptual.

6.1 Moment ideals

Let

$$\mathcal{M}_4 = \{W^i Z^j T^k : i, j, k \geq 0, i + j + k \leq 4\}.$$

This set has 35 elements. Fix a support $S \subset \mathcal{M}_4$, write

$$P_S = \sum_{s \in S} c_s s,$$

and work on the coefficient torus $\prod_{s \in S} c_s \neq 0$. For $s = (i_s, j_s, k_s)$ and a multiplicity vector $\nu = (\nu_s)_{s \in S} \in \mathbb{N}^S$, put

$$|\nu| = \sum_s \nu_s, \quad I(\nu) = \sum_s \nu_s i_s, \quad J(\nu) = \sum_s \nu_s j_s, \quad K(\nu) = \sum_s \nu_s k_s.$$

Wick contraction gives the exact moment polynomial

$$\mathbb{E}(P_S^m) = \sum_{\substack{|\nu|=m, \\ I(\nu)=J(\nu) \\ K(\nu) \text{ even}}} \frac{m!}{\prod_s \nu_s!} I(\nu)! (K(\nu) - 1)!! \prod_s c_s^{\nu_s}. \quad (9)$$

Thus the truncated full-support coefficient scheme is defined by

$$I_{S,M} = \langle \mathbb{E}(P_S), \dots, \mathbb{E}(P_S^M) \rangle : \left(\prod_{s \in S} c_s \right)^\infty. \quad (10)$$

The equations are homogeneous, so one coefficient may be normalized to one before saturation.

There are

$$\sum_{r=1}^4 \binom{35}{r} = 59,535$$

raw supports. We identify a support with its reflection under $W \leftrightarrow Z$. Two elementary filters precede the Groebner calculation. If some moment in (9) has exactly one coefficient monomial, the coefficient torus contains no solution. If no contraction occurs, all charges $i_s - j_s$ have one strict sign and the mixed-moment conclusion follows by bounded weight.

For completeness, the latter test is finite in this range. A zero charge contracts in one or two copies, according to its T -parity. Opposite nonzero charges $q > 0 > r$, with $|q|, |r| \leq 4$, have a primitive zero-sum relation of length

$$\frac{q + |r|}{\gcd(q, |r|)} \leq 8;$$

doubling corrects odd T -parity. Hence every support which is not strictly one-sided has a contraction by order sixteen.

6.2 Exact enumeration

After the preceding filters, the four-term stratum contains 5,718 coefficient-torus ideals requiring elimination. Exact Groebner bases over $\mathbb{Q}$ for $I_{S,20}$ leave precisely the supports

$$\begin{aligned} &\{Z, ZT^2, WZ^2, W^3T\}, \\ &\{ZT, Z^2, WT, W^2\}, \\ &\{ZT, ZT^3, WZ^2T, W^3\}. \end{aligned}$$

The first and third acquire the unit ideal after adjoining moments through order twenty-four. The same exact search in the one-, two-, and three-term strata leaves no active one- or two-term family and two three-term families. Including the four-term survivor, the genuine active null families are

$$P_{\text{lin}} = T + aZ + bW, \quad 1 + 2ab = 0, \quad (11)$$

$$P_{\text{sh}} = Z + aWT - \frac{1}{2}a^2W^3, \quad (12)$$

$$P_{\text{pr}} = ZT + aZ^2 + \frac{1}{2a^2}WT - \frac{1}{4a^3}W^2, \quad a \neq 0. \quad (13)$$

The saturated lexicographic bases are triangular. With the first coefficient normalized to one, their nontrivial equations are respectively

$$1 + 2c_1c_2, \quad 2c_2 + c_1^2, \quad 2c_1^2c_2 - 1, \quad c_3 + c_1c_2^2.$$

These formulas both classify the coefficient schemes and make the survivors easy to interpret.

6.3 All-order interpretation of the survivors

The family (11) is an isotropic linear form. A complex orthogonal change takes it to a scalar multiple of Z . A fixed polynomial Q has bounded W -degree, so $\mathbb{E}(QZ^m) = 0$ for all sufficiently large m .

For (12), define

$$D = W\partial_T - T\partial_Z.$$

Then $D(2WZ + T^2) = 0$, $D(W) = 0$, and

$$P_{\text{sh}} = \exp(-aWD)(Z).$$

Gaussian integration by parts gives $\mathbb{E}(DF) = 0$ for every polynomial F , and therefore also $\mathbb{E}(WDF) = 0$. Since WD is locally nilpotent, $\phi = \exp(-aWD)$ is a polynomial automorphism preserving the Gaussian functional. For fixed Q ,

$$\mathbb{E}(QP_{\text{sh}}^m) = \mathbb{E}(\phi^{-1}(Q)Z^m),$$

which vanishes eventually by the same bounded-weight argument.

Finally,

$$P_{\text{pr}} = \left(Z + \frac{1}{2a^2}W \right) \left(T + aZ - \frac{1}{2a}W \right). \quad (14)$$

The second factor is isotropic and orthogonal to the first. After an orthogonal Witt change, (14) is a scalar multiple of TZ . Every P_{pr}^m then has isotropic weight m , which a fixed Q can balance only finitely often. All three families satisfy GMC. Together with the strictly one-sided supports, this proves Theorem 1.2.

6.4 Reproducibility and scope

The enumeration is implemented with exact integer Wick coefficients and Singular's rational Groebner bases. The certificate script enumerates supports rather than reading a stored survivor list, repeats every saturation over $\mathbb{Q}$, verifies the three displayed triangular bases, and checks (14). It is run from the repository root by

```
.venv/bin/python scripts/verify_three_real_gmc_sparse_pareto.py
```

The complete replay takes several minutes on a laptop. Modular runs are used only as optional discovery filters and are not part of the exact certificate.

7 What the argument says

The proof gives a stronger structural statement than the mixed-moment conclusion alone:

$$\mathbb{E}(P^m) = 0 \text{ for all } m \geq 1 \implies \text{all rotational weights of } P \text{ have one strict sign.}$$

Thus vanishing in two variables is not caused by a complicated balance among positive, zero, and negative angular modes. Such a balance always has a lowest radial face, and a prime moment detects it.

This also explains why support-by-support calculations quickly appeared to show rigidity. A three-level support, a star support, or a tree of primitive weight relations is only a finite presentation of the same lower face. Leaf removal is unnecessary: the supporting line exposes all relevant leaves at once.

The proof is genuinely two-dimensional. With one circular pair there is one weight coordinate and one radial order, and the latter is measured by the single factorial $j!$. In higher dimension there are several radial directions, so the one-dimensional isolation step no longer applies. Corollary 5.1 shows that this is not merely a technical gap: counterexamples already occur in dimension three.

8 Formal verification

A companion Lean 4 development has been prepared as supplementary material for the two-variable theorem. It mechanically checks the factorial-quotient divisibility, prime-dilated factorial isolation, commutation of Frobenius with angular constant-term extraction, the final prime contradiction, and the eventual vanishing argument for both signs of the rotational support. The development contains no `sorry` declarations. The computer-assisted sparse theorem is certified separately by the exact rational script described in Section 6.

The formalization exposes three external inputs as named axioms: the Duistermaat–van der Kallen constant-term theorem, rational supporting-face extraction, and the existence of good finite-characteristic specializations of finite-type $\mathbb{Z}$ -domains. Thus the artifact gives a machine-checked verification of the arithmetic and constant-term core of the proof, with its remaining trust boundary stated explicitly.

Acknowledgments and declaration of generative AI use

The author used OpenAI’s ChatGPT and Codex extensively during the research process, including for conjecture formation, symbolic experimentation, proof exploration, consistency checks, and preparation of this manuscript. The lower-face and prime-isolation argument emerged from that collaborative process. The author reviewed the mathematical claims and assumes full responsibility for the contents. The proof of Theorem 1.1 is noncomputational. Theorem 1.2 uses the exhaustive exact coefficient-scheme calculation stated in the text and supplied with the paper.

References

- [1] Harm Derksen, Arno van den Essen, and Wenhua Zhao. The Gaussian Moments Conjecture and the Jacobian Conjecture. *Israel Journal of Mathematics*, 219(2):917–928, 2017. DOI.
- [2] J. J. Duistermaat and W. van der Kallen. Constant terms in powers of a laurent polynomial. *Indagationes Mathematicae*, 9(2):221–231, 1998. DOI.

- [3] Christopher D. Long. Small counterexamples to the Gaussian Moments Conjecture, 2026. arXiv:2607.18186.
- [4] The Stacks Project Authors. Chevalley's theorem. The Stacks Project, Tag 00FE, 2026.
- [5] The Stacks Project Authors. Constructible neighbourhoods of generic points. The Stacks Project, Tag 0AAW, 2026.